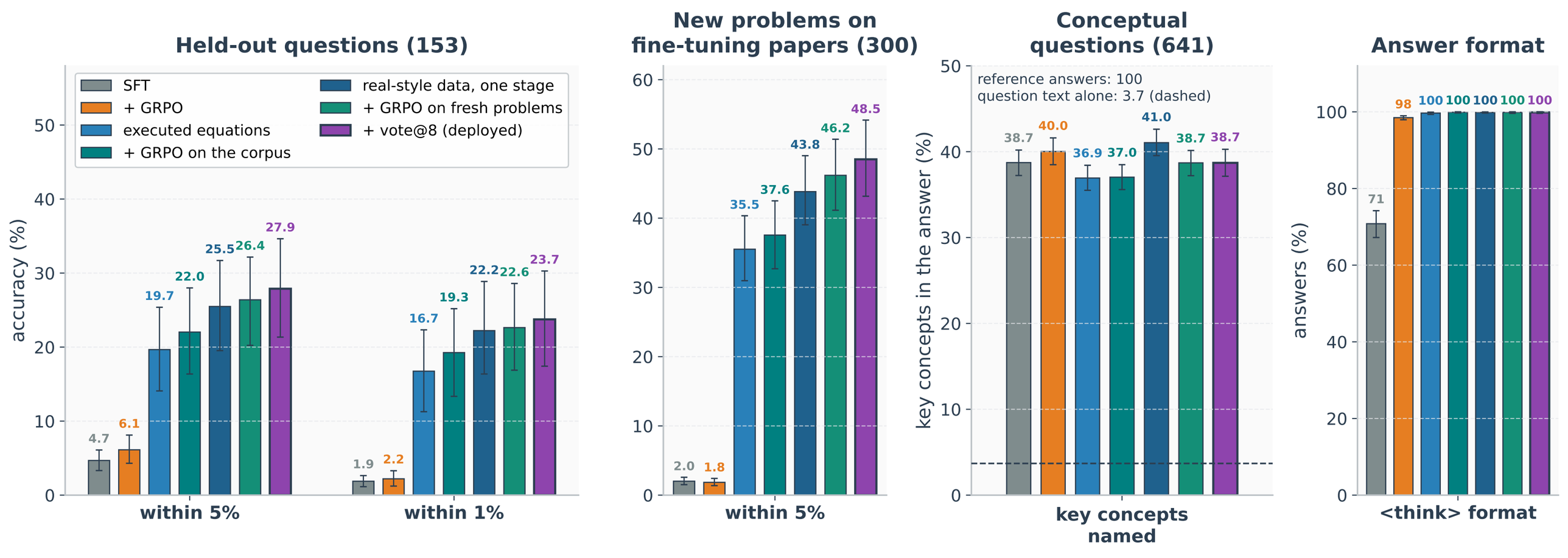
Held-out questions (153)
SFT
+ GRPO
executed equations
+ GRPO on the corpus
real-style data, one stage
+ GRPO on fresh problems
+ vote@8 (deployed)
accuracy (%)
4.7
6.1
19.7
22.0
25.5
26.4
27.9
1.9
2.2
16.7
19.3
22.2
22.6
23.7
within 5%
within 1%
New problems on fine-tuning papers (300)
2.0
1.8
35.5
37.6
43.8
46.2
48.5
within 5%
Conceptual questions (641)
reference answers: 100
question text alone: 3.7 (dashed)
key concepts in the answer (%)
38.7
40.0
36.9
37.0
41.0
38.7
38.7
key concepts named
Answer format
answers (%)
71
98
100
100
100
100
100
<think> format